S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.
(An Autonomous Institute, Affiliated to RTMNU, Nagpur)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
"To become a center for quality education in the field of computer science & engineering and to create competent professionals."

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TAE 1: Activity Based Learning

A Case Study

TITLE: Data Shaping and Transformation (Merge, Group By, Pivot) Using Power BI

Submitted By

Prathamesh Deshmukh

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Course Instructor Signature: _____

1. Title

Data Shaping and Transformation (Merge, Group By, Pivot) Using Power BI

2. Content

This report focuses on transforming and restructuring data using Power BI to make it suitable for analysis and reporting.

Initially, data from multiple sources was combined using the **Merge Queries** feature. The main dataset was merged with the Region Mapping table based on the Region column to add relevant information such as the Manager.

After merging, data was summarized using the **Group By** operation. The dataset was grouped based on Category, and total revenue was calculated by summing the Sales values.

Further, the structure of the data was modified using the **Pivot Column** feature. The Status column was transformed into multiple columns such as Shipped, Pending, and Returned, each representing sales values for better comparison and analysis.

Finally, the **Query Dependencies** view was used to visualize how different datasets are connected and transformed into the final output. This helped in understanding the data flow and relationships between queries.

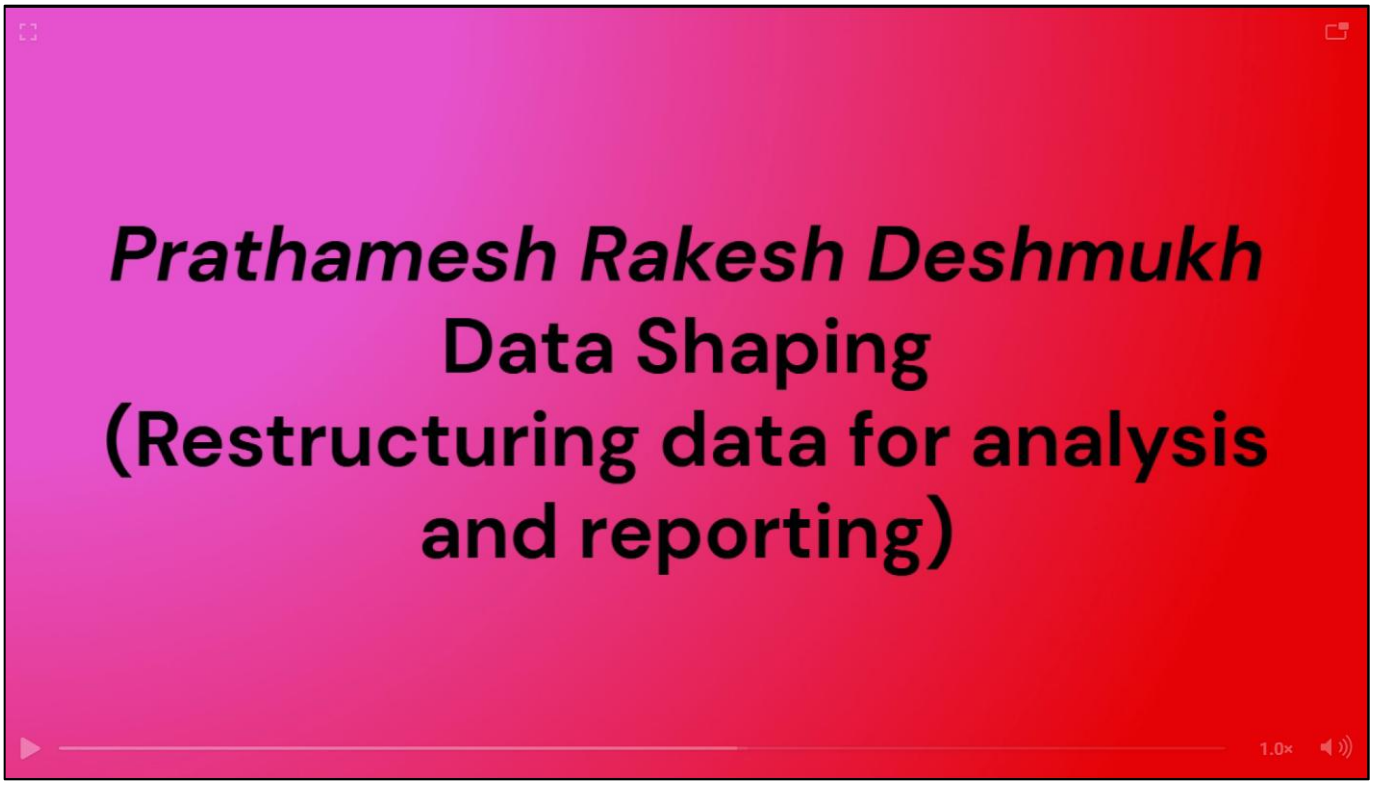
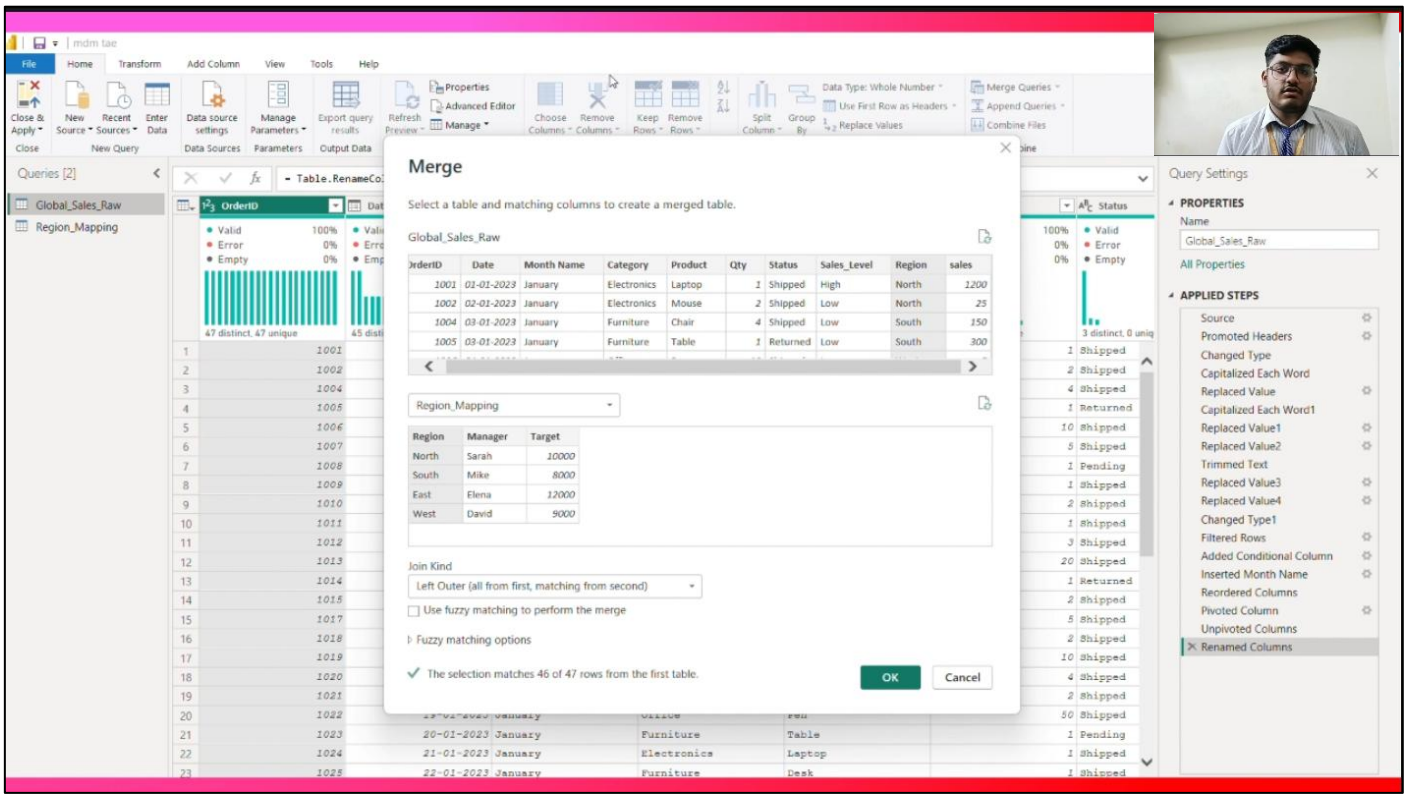
3. Purpose

• To combine data from multiple sources
• To summarize and aggregate data
• To restructure data for better analysis
• To understand data flow using query dependencies

4. Method

• Merged main dataset with Region Mapping table
• Expanded merged column to include Manager
• Grouped data by Category and calculated total revenue
• Pivoted Status column into multiple columns
• Analyzed query dependencies for data flow

5. Conclusion

Data shaping is an important step in data preparation. By merging datasets, summarizing values, and restructuring data, it becomes easier to analyze and visualize information. These transformations improve data usability and support effective decision-making.

Video Screenshot: